

Topic 6: Lists

Summary & Learning Outcomes

Bengkel ToT : Chapter 6

An overview of Topic 6

Learning Outcomes (1 of 2)

- **6.1** Identify the concept and basic structure of lists in Python.
- **6.2** Explain element indexing and basic list operations in Python.
- **6.3** Apply lists with indexing, assignment statements, and for loops to solve simple problems.
- **6.4** Identify pre-defined list functions (`len()`, `min()`, `max()`, `sum()`) and a method (`append()`).
- **6.5** Apply pre-defined list functions (`len()`, `min()`, `max()`, `sum()`) and the method `append()` to process list data.

Learning Outcomes (2 of 2)

- **6.6** Construct programs that perform list method (`sort()`) to sort list data. (*Assignment*)
- **6.7** Determine output from list data using control structures.
- **6.8** Construct Python programs using `len()`, `min()`, `max()`, `sum()` and `append()`.
- **6.9** Apply lists with control structures to develop simple Python programs.

Content Summary

The topic is covered in six parts:

1. **What is a list** : storing many values in one variable
2. **Indexing** : accessing elements by position
3. **Looping** : going through a list
4. **List operations** : creating and updating
5. **Functions & methods** : `len` , `min` , `max` , `sum` , `append`
6. **Sorting** : ordering a list with `sort()`

1. What Is a List?

- **Definition** : storing many values in one variable
- **Syntax** : square brackets and elements
- **Data types** : same type, mixed type, empty list
- **Characteristics** : ordered, mutable, duplicates, dynamic size

2. Indexing

- **Positive index** : position from the front
- **Negative index** : position from the back
- **IndexError** : accessing a position that does not exist

3. Looping Through a List

- **For-each loop** : visiting every element
- **For loop with index** : working with positions
- **While loop** : looping with a condition

4. List Operations

- **Creating** : empty or with elements
- **Updating** : changing an element by its index

5. Built-in Functions & Methods

- Functions : `len()`, `min()`, `max()`, `sum()`
- Method : `append()`
- Function vs method : the difference in how each is written

6. Sorting

- `sort()` : the sorting method
- **Ascending / descending** : order direction
- **Numbers and strings** : what can be sorted

End

Our [Chapter 6 ToT Lab](#) puts each of these outcomes into practice, hands-on.